Efficiency and Wildness Are Jointly Produced: A Three-Way Refutation in Minimal Market Models

Abstract

Agent-based models reproduce financial stylized facts piecemeal; whether a single minimal mechanism can buy them all is open. We score minimal flow-based markets against a ten-event battery calibrated so that real equity data passes all ten events and geometric Brownian motion exactly three. A market of fundamentalists, chartists, volatility-targeters and market makers reaches five of ten, and its failures are diagnostic: the missing events concern long memory, time-irreversibility, and information-free price signs. We then test, in three pre-registered experiments with kill criteria, whether rationality can close the gap: (i) an anticipatory agent that front-runs the forecastable flow; (ii) the fixed point of mutual anticipation, iterated to depth five; (iii) a price-setting maker that provably absorbs the forecastable component into the price level. All three fail, and they fail informatively: deeper anticipation *grows* the sign leak while contracting the flow it targets, and exact absorption kills the fat tails, the leverage effect and volatility clustering along with the leak. We conclude that in this model class, sign-efficiency and the wild stylized facts are jointly produced by the same state-riding flows and are not separately purchasable. An empirical counterpart — the leverage effect's sign ordered by market microstructure across equities (-0.04), FX (≈ 0), and crypto ($+0.03$) — is consistent with the mechanism. All results are pre-registered, gate-verified against synthetic ground truth, and regenerable from the public repository.

Keywords: stylized facts; agent-based markets; market efficiency; price impact; leverage effect; pre-registered negative results.

1 Introduction

Real price series manage two things at once: their return signs are nearly unforecastable, and their amplitudes are wild — heavy tails, clustered volatility, crash asymmetry. This paper asks whether a minimal market mechanism can buy both properties together, and answers by refutation: in flow-based minimal markets, sign-efficiency and wildness are jointly produced by the same flows, and three pre-registered attempts to purchase them separately — rationality added as a trading agent, as the fixed point of mutual anticipation, and as a price-setting rule — all fail against kill criteria declared in advance. The refutations are the result: they compose into a structural claim about what this model class cannot do, measured on a fixed instrument (Figure 1).

The question sits in a literature with a shared target list: the stylized facts — heavy tails, volatility clustering, the leverage effect, near-absence of linear return autocorrelation — that any candidate mechanism for price formation should reproduce (Cont, 2001). Agent-based modeling took up the challenge early: heterogeneous-agent markets with fundamentalists and

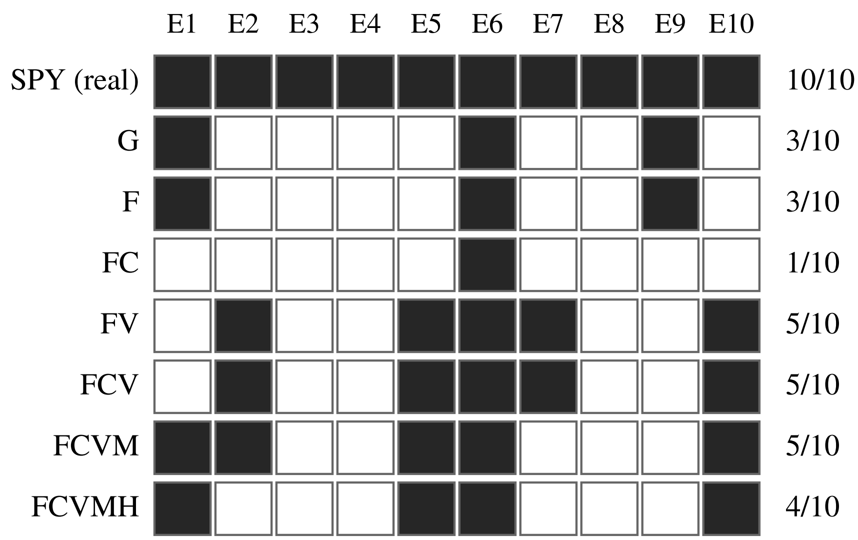

Figure 1. Battery scorecard: real SPY against the ablation ladder (filled = pass). G = noise only (GBM); F adds fundamentalists; C chartists; V volatility-targeters; M market makers; H multi-horizon heterogeneity. No flow configuration exceeds 5/10, and the five events missing from the best configuration (FCVM) are E3, E4, E7, E8, E9.

chartists generate fat tails and clustered volatility from agent interaction rather than from exogenous news (Lux and Marchesi, 1999), and a substantial literature has since mapped which ingredients produce which facts (LeBaron, 2006). Two gaps in that literature motivate the design here. First, scoring is usually piecemeal: a model is exhibited that matches two or three facts, with the remaining facts unexamined, so the literature accumulates existence proofs rather than a ledger of what a given mechanism class can and cannot buy. Second, negative results are structurally underproduced: when an added ingredient fails to improve a model, the failure is rarely reported, almost never pre-registered, and so carries no evidential weight. We address both gaps with a fixed measurement instrument and a pre-registered refutation chain.

Contribution (i): a battery, calibrated at both ends. We distill ten events (E1–E10) from a prior measurement program on real markets — price efficiency, fat tails, volatility clustering, long memory in the volatility clock, the leverage effect, conditional Gaussianity given the volatility path, clock jumps, the arrow of time in the return–volatility coupling, information-free price signs, and gain/loss tail asymmetry — each with a numeric pass criterion fixed before any simulation was run (Section 2). The battery is calibrated so that real SPY data passes 10/10 while geometric Brownian motion passes exactly 3/10; this spread is the instrument's dynamic range, and it is enforced by an automated gate that runs on every commit.

Contribution (ii): a pre-registered refutation arc. The best flow-only configuration of a minimal market stalls at 5/10, and the five missing events are diagnostic: they concern long memory, time-irreversibility, and the information content of price signs. Three successive experiments — each pre-registered with hypotheses, budgets, and kill criteria before any evaluation run — then test whether *rationality* can close the gap: a single anticipatory agent (Section 3), the fixed point of mutual anticipation iterated to depth five (Section 4), and a price-setting market maker that absorbs the forecastable flow into the price level by algebraic identity (Section 5). All three are refuted, and because each refutation was armed with a mechanism gate proving the tested agent *works*, the refutations compose into a structural claim (Section 6): in this model class, sign-efficiency and the wild stylized facts are jointly produced by the same state-riding flows and cannot be purchased separately.

Contribution (iii): a methodology that makes negative results evidential. Every estimator is licensed by a convict-and-exonerate gate — it must detect a planted effect on synthetic data where the effect exists and must stay silent where it does not — before it touches evaluation data (Section 9). Every experiment pre-declares its parameter budget, its contingent tuning pass, and its tie-break rule; the third experiment additionally pre-declared a closure clause naming itself the final attempt on the line. Under this regime a refutation is not an absence of result; it is a measurement.

The headline mechanism result is easy to state. Adding one layer of anticipation to the minimal market leaves its score unchanged and moves the sign leak one derivative earlier; iterating anticipation five deep contracts the model-space forecastable flow by a factor of four while the leaked sign information in the realized prices moves the wrong way (median direction bits $0.0184 \to 0.0200 \to 0.0257$; the depth-five rise holds on 7 of 8 paired seeds); and a maker that removes the forecastable component's price impact exactly — verified to the identity on a toy world — does not close the leak either, but instead deletes the fat tails (kurtosis $8.8 \to 3.0$), the leverage effect ($-0.125 \to -0.008$), and volatility clustering along with the wildness the flows were generating. Efficiency in signs and wildness in amplitudes are not separate dials in this model class. Section 7 adds an empirical observation that is consistent with the mechanism — the leverage effect's sign across equities, FX, and crypto is ordered exactly as the flow-structure story predicts — without claiming it as proof.

2 The battery and the base market

2.1 Ten events

The battery consists of ten binary events, computed identically on real data and on simulated output, with pass criteria fixed before any agent-based run. Table 1 lists them; Appendix A gives the exact statistic definitions and thresholds as implemented, together with the full statistic vector at both calibration ends. The events are close-only versions of regularities measured and gate-validated in a prior program on ∼48 liquid US equities and ETFs (2010–2026), so the thresholds are empirical ranges, not conventions.

Calibration pins both ends of the scale: real SPY (close-only) scores 10/10; a pure GBM scores exactly 3/10, passing only E1, E6 and E9 — Gaussian worlds are efficient, trivially one-clock, and carry no sign information. Everything between 3 and 10 is what a mechanism must buy. Two scoping remarks. The thresholds are ranges from the multi-asset program, but the joint-10/10 anchor is a single index; the battery's job in this paper is comparative — every configuration is scored on the same events with the same seeds — not to certify any absolute realism. And a battery score carries no formal standard error; what protects the comparisons is that they are paired (all configurations share seeds 100–107) and that every conclusion below rests on a pre-registered kill criterion, not on a one-event margin.

#	Event	Pass criterion
E1	price efficiency	$\mathrm{AC}_1(r) \in [-0.15, 0.05]$ (allows the real index's mild daily reversal)
E2	fat tails	$\mathrm{kurtosis}(r) \in [4.5, 40]$
E3	vol clustering	$\mathrm{AC}_1(\lvert r\rvert) \geq 0.12$ and mean $\mathrm{AC}(\lvert r\rvert, \text{lags } 5\text{–}20) \geq 0.05$
E4	long-memory clock	8-week volatility-clock level autocorrelation above threshold
E5	leverage effect	mean $\mathrm{corr}(r_t, \mathrm{RV}_{t+1..10}) \leq -0.03$
E6	one-clock Gaussianization	$\mathrm{kurtosis}(r/\sigma_{5d}) \leq 5$
E7	clock jumps	AR(1) clock-innovation skew beyond the log-χ^2 noise floor
E8	arrow of time in the coupling	entropy production significant in r and reduced $\geq 25\%$ by vol deformation
E9	no sign information	direction bits $\leq$ shuffle null (95%)
E10	tail asymmetry	gain/loss tail asymmetry at 2.5σ

Table 1. The ten-event battery (final calibrated form; exact estimator definitions and thresholds in Appendix A). E1, E4, E7 and E10 were recalibrated against real data before any ablation was run, under a gate requiring real SPY to score 10/10 and geometric Brownian motion exactly 3/10; notably, a GJR-GARCH process fails E4 — exponential memory cannot fake long memory. E9 measures the mutual information between three public tape features and the next-day return sign with a discrete plug-in estimator against a permutation null.

2.2 The minimal market

The base market is deliberately small: aggregated flows, about a dozen parameters, linear impact. Log price p_t moves by

$$r_t \;=\; \lambda \left[D_t^{\mathrm{fund}} + D_t^{\mathrm{chart}} + D_t^{\mathrm{vol}} + D_t^{\mathrm{mm}} + D_t^{\mathrm{noise}} \right], \tag{1}$$

with $D^{\mathrm{fund}} = k_F\,(V_t - p_t)$ for a random-walk log fundamental V; $D^{\mathrm{chart}} = k_C \tanh(m_t/s)$ for an EWMA trend signal m; $D^{\mathrm{vol}} = k_V\,(L_t - L_{t-1})$ for volatility-targeters (hereafter vol-targeters) running the leverage rule $L = \min(L_{\max}, \sigma^*/\hat{\sigma})$ against an EWMA variance estimate $\hat{\sigma}^2$; a one-lag market-maker flow $D^{\mathrm{mm}} = -k_M\, r_{t-1}$; and i.i.d. noise. The volatility-targeting flow is the reflexive ingredient: a vol spike forces de-leveraging (selling), which moves price, which raises vol. The market-maker ingredient was added at the calibration stage, before any ablation was scored, when tuning showed that vol-targeting flows structurally leak momentum — their de-leveraging is serially correlated — and that in reality such flow is absorbed by liquidity provision.

Parameters were hand-set once on the full configuration to produce sane volatility levels and pass E1–E3 only, and were not tuned per configuration or per event; ablations zero out flows and change nothing else. Eight seeds per configuration, $T = 6000$; an event passes if a majority of seeds pass.

2.3 The ablation ladder and the 5/10 ceiling

Figure 1 shows the ladder. Noise alone (G) and fundamentalists alone (F) score the Gaussian 3/10. Chartists without discipline are toxic: FC scores 1/10, because the momentum leak

	$AC_1(r)$	kurtosis	$AC_1(\lvert r\rvert)$	leverage	E9 bits
SPY (real)	-0.103	15.1	0.31	-0.113	0.0029
GBM (G)	0.001	3.0	-0.001	0.007	0.0001
FCVM (best flow-only)	0.036	8.8	0.38	-0.125	0.0184

Table 2. Key medians. FCVM reproduces efficiency, tails, clustering amplitude and the leverage effect at realistic magnitudes; what it cannot do is keep the sign channel silent (E9: 0.0184 significant direction bits against SPY's 0.0029) or generate long memory and the coupling arrow.

destroys even efficiency. The vol-targeters are the star ingredient: FV jumps to 5/10, buying the wild one-sided facts — fat tails (E2), the leverage effect (E5), clock jumps (E7), crash asymmetry (E10) — because the de-leveraging spiral is the only flow that reacts to volatility and reacts in one direction only. Adding market makers (FCVM) restores efficiency (E1) at the cost of E7, holding the score at 5/10: efficiency and wildness are contributed by *different* agents. Multi-horizon heterogeneity (FCVMH) fails both ways, dropping to 4/10.

The ceiling is 5/10 and the five missing events are a diagnosis (Table 2). FCVM fails E3 (slow volatility clustering), E4 (long-memory clock), E7 (clock jumps, marginally), E8 (the arrow in the coupling), and E9 (information-free signs). The E9 failure is the paper's through-line, so we give it a name used throughout: the *sign leak* — a significant excess of E9's statistic, the *direction bits* $I(\text{tape features}; \operatorname{sign} r_{t+1})$, over its shuffle null. FCVM leaks 0.0184 bits where real SPY shows 0.0029. The leak has a known source: the vol-targeters' de-/re-leveraging drift is serially correlated and forecastable from the tape, and a one-lag market maker cannot absorb a multi-day drift. The natural hypothesis — the one this paper's arc tests to destruction — is that the missing organ is *expectation*: rational agents who anticipate the predictable flow should arbitrage the sign leak away and perhaps stretch the volatility episodes into long memory.

3 Experiment I: one anticipator

3.1 Design

The vol-targeters' flow is mechanical and forecastable from public information alone: their mandate ($L = \min(L_{\max}, \sigma^*/\hat\sigma)$) and their estimator (an EWMA of squared returns at a known speed) are both computable from the price tape. The anticipatory agent A is handed exactly that and nothing else. At each step it reconstructs $\hat\sigma_t^2$ from the return history, forecasts the integrated future mechanical flow under the one belief that defines vol-targeting — volatility reverts to the targeters' own target, so leverage ends at $L_{\text{eq}} = \min(L_{\max}, 1)$ —

$$\hat F_t \;=\; k_V \cdot \operatorname{mean}_{\text{cohorts}}\big(L_{\text{eq}} - L_t\big), \tag{2}$$

holds inventory proportional to the forecast, capped by capital, $I_t^* = \operatorname{clip}(k_A \hat F_t, \pm\text{cap}_A)$, and trades the inventory change plus execution noise. The unwind is automatic: as vol reverts and the targeters re-lever, the forecast decays and the anticipator sells into exactly that mechanical bid. Three parameters (k_A, cap_A, s_A), frozen before any battery run.

Three gates license the mechanism before any score is read (the pattern is described in Section 9): (a) with the agent disabled the simulator reproduces the original market byte-for-byte, protecting the SPY-10/GBM-3 calibration; (b) the forecast is causal — tampering with future returns leaves the trade prefix unchanged; (c) on a deterministic toy world with a

Config	Score	Failed events	E9 bits (median)
FCVM (control)	5/10	E3 E4 E7 E8 E9	0.0184
FCVM+A (hypothesis)	5/10	E3 E4 E7 E8 E9	0.0200
FV+A	5/10	E1 E3 E4 E8 E9	0.0158
F+A (no targeters)	3/10	= F exactly	0.0002

Table 3. Experiment I ablation (8 evaluation seeds, majority vote). Adding the anticipator changes nothing that was supposed to change: the failure set is identical and the direction bits do not fall.

perfectly predictable mechanical flow, the anticipator profits *and* damps the flow's price impact, from 1.00 to 0.75 per unit of arriving flow. The agent demonstrably works. The pre-registered kill criterion: if the frozen configuration does not exceed 5/10 on the evaluation seeds, the expectation hypothesis is refuted.

The parameter protocol matters for what follows. The first-shot parameters $(k_A, \mathrm{cap}_A, s_A) = (0.5, 0.02, 0.002)$ scored 4/10 on tuning seeds disjoint from evaluation, so the single pre-registered grid pass was spent: its best score was 5/10, tied across all six $k_A = 0.25$ combinations plus one weak $k_A = 0.5$ setting — flat wherever the anticipator is weak. The pre-declared deterministic tie-break (lexicographically smallest, i.e. the weakest anticipator) selected $(0.25, 0.01, 0.001)$. That the selection lands on "trade as little as possible" was recorded, at registration time, as already being evidence about the hypothesis.

3.2 Result: refuted, with the leak moved one derivative earlier

Table 3 shows the outcome. The kill criterion fires: FCVM+A scores 5/10 with a failure set identical to the control's. The events expectation was supposed to buy did not move — the direction bits did not move toward zero (medians $0.0184 \to 0.0200$; the paired per-seed comparison splits 4–4, so the licensed reading at one layer is "unchanged," not "reduced"), and the long-memory events stayed flat. The interaction hypothesis held: without vol-targeters the anticipator buys nothing (F+A = F = 3/10) — expectation needs a predictable flow to trade against; it simply does not fix anything when it has one.

Why it fails is the finding. The tuning grid is monotone: the harder the anticipator trades, the worse the market scores. Efficiency breaks (its unwind adds reversal: AC_1 falls from -0.09 toward -0.35 as k_A rises), the tails erode (it damps the very spiral that generates them, kurtosis $8.8 \to 4.6$), and the sign leak *grows* (up to 0.039 bits at full strength $k_A = 1$). The mechanism is structural, not parametric: the anticipator's inventory is a function of the same slow public volatility state as the flow it front-runs, so its own future unwind is exactly as forecastable as what it absorbs. One layer of expectation does not remove the predictable flow; it reproduces the leak one derivative earlier, while eating the wildness. The refutation names its own successor: if one anticipator re-leaks because its inventory is public-state-driven, perhaps information-free prices require anticipators who anticipate *each other* — the fixed point of mutual anticipation.

Config	Score	Failed events	E9 bits (median)
K=0 (FCVM, control)	5/10	E3 E4 E7 E8 E9	0.0184
K=1 (Exp. I layer, byte-identical)	5/10	E3 E4 E7 E8 E9	0.0200
K=5 (frozen carry-over)	3/10	E1 E3 E4 E5 E7 E8 E9	0.0257
K=5 (tuned, contingent pass)	5/10	E3 E4 E7 E8 E9	0.0168

Table 4. Experiment II: the K-ladder. The median sign leak grows in K — the depth-five rise holds on 7 of 8 paired seeds — while the model-space forecastable flow contracts toward zero; the frozen K=5 stack is a two-event regression (it breaks efficiency and kills the leverage effect).

4 Experiment II: the fixed point of mutual anticipation

4.1 The iterated operator and its algebra

The fixed-point stack makes each layer's unwind part of what is anticipated. Layer k front-runs the *residual* forecastable total flow — the mechanical flow plus the deterministic future unwind of the $k-1$ layers beneath it:

$$\mathrm{resid}_0 = \hat{F}, \qquad J_k = \mathrm{clip}\big(k_A\,\mathrm{resid}_{k-1}, \pm\mathrm{cap}_A\big), \qquad \mathrm{resid}_k = \mathrm{resid}_{k-1} - J_k, \qquad I_K = \textstyle\sum_{k=1}^{K} J_k. \tag{3}$$

Away from the caps this telescopes: $I_K = \big(1-(1-k_A)^K\big)\hat{F}$, and the model-implied forecastable residual of total flow contracts geometrically, $\mathrm{resid}_K/\hat{F} = (1-k_A)^K \to 0$. $K=0$ is the flow-only control; $K=1$ is byte-for-byte Experiment I's single layer; $K\to\infty$ is the rational-expectations limit of this operator class. The budget was fixed before any run: $K=5$ exactly, no convergence-based stopping, evaluation on the same eight seeds, and a contingent six-candidate tuning pass permitted only on a regression below the control's 5/10, with the same weakest-anticipation tie-break.

One amendment was made at the gate stage, before any battery run, and it illustrates what the gates are for: the first formulation capped the *stack's* total inventory at a single agent's cap_A, and the pre-specified mechanism gate failed it — under a shared cap the K=5 stack holds exactly the K=1 inventory through every large episode and the measured forecastable-flow contraction disappears. A total-cap stack cannot express the hypothesis it exists to test. The licensed formulation caps each layer separately.

With per-layer caps the mechanism gate passes: on the deterministic toy world the projection of realized total flow onto the mechanical flow contracts as $\beta_K = 1.000 \to 0.750 \to 0.237$ at $K = 0 \to 1 \to 5$, matching the $(1-k_A)^K$ theory line exactly. The operator does what the rational-expectations argument says it should. The question is whether the market obeys the argument.

4.2 Result: the inversion

Table 4 and Figure 2 give the result, which is the paper's centerpiece. The kill criterion fires again, and this time the two diagnostics move in *opposite directions*: as K goes $0 \to 1 \to 5$, the model-space forecastable flow falls $1.000 \to 0.750 \to 0.237$ while the market-space sign leak rises $0.0184 \to 0.0200 \to 0.0257$ bits in the medians, with K=5 above the control on 7 of 8 paired seeds. The frozen K=5 stack is moreover a two-event regression: its aggregate unwind adds reversal (AC_1: $-0.075 \to -0.247$, breaking E1) and it absorbs the very spiral that generates

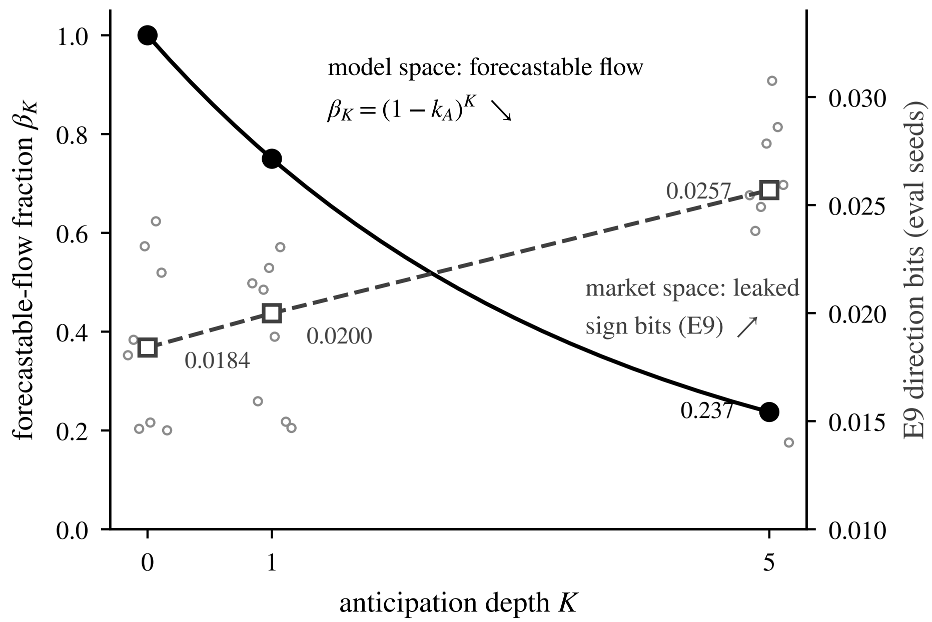

Figure 2. The inversion. Solid curve and filled circles (left axis): the forecastable-flow fraction β_K measured on the deterministic toy world, falling exactly as $(1-k_A)^K$ — 1.000 → 0.750 → 0.237. Dashed line and open squares (right axis): the E9 direction bits measured in the simulated market at the same K (open circles: the eight evaluation seeds), medians rising 0.0184 → 0.0200 → 0.0257; the K=5 rise holds on 7 of 8 paired seeds. The operator removes forecastable flow in model space; the market re-creates sign information in price space. Open-loop contraction does not survive closed-loop equilibrium.

the leverage effect (E5 dies, $-0.095 \to -0.030$), while tails erode (kurtosis 6.8 → 5.3) and crash asymmetry collapses (tail ratio 44.8 → 4.3). Rational front-running at depth does not merely fail to buy the missing events; at carry-over strength it kills wild facts the market already had.

The regression triggered the pre-registered contingent pass: six candidates on disjoint tuning seeds came back 5/10 at best, tied across the three weakest settings, and the pre-declared tie-break again selected the weakest — $k_A = 0.05$, an effective stack strength of $1 - 0.95^5 = 0.226$, *weaker than Experiment I's single frozen layer* (0.25). Read once on the evaluation seeds: 5/10, failing exactly the control's five events. The selection pattern of Experiment I repeats one level up: the battery pays for less anticipation at every opportunity, never more.

The algebra explains the failure sharply. Away from the caps, the K-layer stack is equivalent to *one* anticipator of strength $1 - (1 - k_A)^K$: mutual anticipation of this operator class deepens into a stronger single anticipator, and Experiment I already measured that stronger is worse. The contraction that defines the fixed point is an open-loop property, licensed by the gate against an exogenous tape; it does not survive closed-loop equilibrium, because every layer's inventory is a function of the same slow public volatility state, so the stack's own price impact regenerates exactly the forecastable structure the model says it removes. Model-space forecastable flow down, market-space sign leak up: this is the cleanest statement we know of why rational-expectations arguments cannot be reached by stacking boundedly-capitalized flow traders.

Config	Score	Failed events	E9 bits (median)
FCVM (control)	5/10	E3 E4 E7 E8 E9	0.0184
FCVM+Q(λ_Q=1.0) (theory case)	1/10	all but E6	0.0237
FCVM+Q(λ_Q=0.5)	3/10	E1 E2 E3 E4 E7 E8 E9	0.0180
Q(λ_Q=0.05) (tuned, contingent pass)	5/10	E3 E4 E7 E8 E9	0.0176

Table 5. Experiment III. Full skew removes the vol-targeters' forecastable impact exactly, and the sign leak does not close — it grows (0.0237, higher than the control on 7 of 8 seeds) — while the wild facts die.

5 Experiment III: price-setting rationality

5.1 The quote-skewing maker

Experiments I and II killed every agent that *trades* against the forecastable flow: an anticipator's inventory rides the same slow public state, so its trades re-leak what they absorb. One hypothesis survives that diagnosis: price-*setting* rationality — a market maker who does not trade but quotes, pre-adjusting the price by the expected impact of the forecastable flow so that the information is absorbed into the price *level* rather than left in the return. The maker reuses the same causal public-state forecast $\hat{F}_t$ of Eq. (2), holds no inventory, posts no flow, and draws no noise:

$$q_t = \lambda_Q \, \lambda \, \hat{F}_t, \qquad r_t = \lambda \, D_t + (q_t - q_{t-1}), \tag{4}$$

with skew strength $\lambda_Q \in \{1.0, 0.5\}$ fixed in advance ($\lambda_Q = 0$ is byte-identical to the unmodified simulator). Because the maker's forecast state *is* the targeters' public state, the quote revision telescopes exactly against the mechanical flow, $q_t - q_{t-1} = -\lambda_Q \, \lambda \, f_t^{\text{mech}}$: at full skew the mechanical flow's entire price impact is absorbed into the level by algebraic identity, not by tuning, and the realized return keeps only the unforecastable flow surprise. Nothing is left that could re-leak — no inventory, no unwind, no cap, no execution noise. The flow itself still executes untouched; only where its impact lands (level versus return) changes.

The mechanism gate verifies this to the identity on the deterministic toy world: at $\lambda_Q = 1$ the leak estimator $\text{corr}(f_t^{\text{mech}}, r_{t+1})$ collapses from $+0.66$ to -0.06 while the mechanical-flow series itself is essentially unchanged — absorption into the level, not suppression of the flow. The registration declared this the third and final attempt on the expectation line and included a closure clause: if the star hypothesis is refuted at this budget, the 5/10 ceiling is declared structural for this model class and the line closes.

5.2 Result: absorption deletes the wildness, not the leak

Table 5 and Figure 3 give the sharpest refutation of the three. The full-skew maker removes the vol-targeters' forecastable impact exactly — and the sign leak does not close. Bits stay flat at half skew (0.0180 versus 0.0184) and *grow* at full skew (0.0237, higher than the control on 7 of 8 seeds). The leak was never the vol-targeters' drift alone: Experiment I's original attribution is overturned. It lives distributed across *all* the state-riding flows — chartist momentum, fundamentalist reversion, the one-lag maker's partial absorption — and deleting the spiral's impact strips away the wildness that was masking them: efficiency breaks (AC_1: $+0.04 \to -0.26$; with the spiral gone the liquidity provider's reversion dominates), and the newly-Gaussian returns make the remaining structure easier to detect.

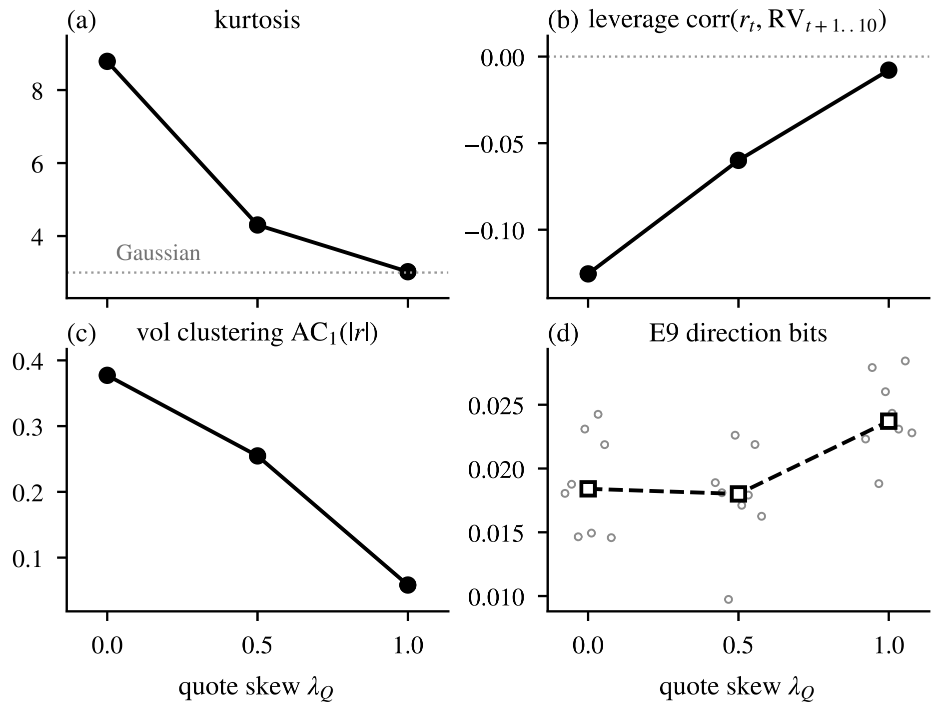

Figure 3. Wild-fact medians versus quote skew λ_Q (8 evaluation seeds). Full absorption of the forecastable flow's price impact kills the fat tails (kurtosis $8.8 \to 3.0$), the leverage effect ($-0.125 \to -0.008$) and volatility clustering ($\mathrm{AC}_1(|r|)$ $0.38 \to 0.06$) — while the E9 direction bits (panel d; open circles are seeds) do not close: flat at half skew, higher at full skew.

The second pre-registered hypothesis — that the maker would neutralize only the forecastable *mean* of the flow, leaving the volatility dynamics intact — is also refuted at the theory case, and its refutation is the paper's central fact: in this market the forecastable flow *is* the wildness generator. Full absorption kills the fat tails (kurtosis $8.8 \to 3.0$), the leverage effect ($-0.125 \to -0.008$), crash asymmetry (tail ratio $40 \to 1.1$) and volatility clustering ($\mathrm{AC}_1(|r|)$ $0.38 \to 0.06$) in one stroke. Half skew regresses too (3/10), so the contingent pass fired: the six-candidate ladder came back flat at 5/10 for every $\lambda_Q \leq 0.30$, and the pre-declared weakest-skew tie-break picked $\lambda_Q = 0.05$ — the same selection pattern a third time, now for a pricing rule. Read once on the evaluation seeds: 5/10, failing exactly the control's five events. The battery pays for the least rationality at every opportunity, in every agent class.

Figure 4 collects the E9 evidence across all three experiments.

6 The structural claim and its limits

6.1 The closure clause as methodology

Experiment III's registration contained, alongside its hypotheses and budget, a closure clause: this is the third and final attempt on the expectation line; if the star hypothesis is refuted at this budget, the 5/10 ceiling is declared structural for flow-based minimal markets with boundedly-rational agents of every class here considered, and the line closes — no fourth experiment. The clause is not rhetoric; it is the instrument that converts three negative results

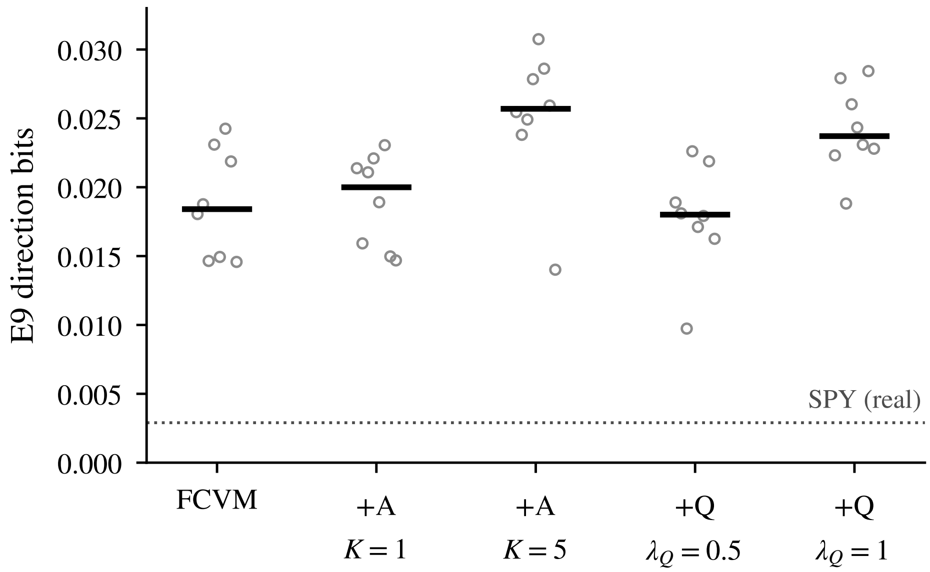

Figure 4. E9 direction bits across every rationality configuration (open circles: evaluation seeds; bars: medians). No configuration approaches the information-free regime; real SPY (dotted, 0.0029) sits far below all of them. Trading against the flow, iterating that trade to depth five, and absorbing the flow's impact into the price level all leave the sign channel open or wider.

into one positive claim. An open-ended sequence of failed attempts supports only "we have not yet found the right agent." A pre-committed final attempt, refuted, supports a stronger statement about the class that was searched — precisely because the search was budgeted, the budget was spent, and the stopping rule was declared before the last result was known.

6.2 The claim

The three refutations compose. In this model class, the forecastable flow component and the wild one-sided facts are the same object:

1. an agent who *trades* against the forecastable flow adds its own forecastable flow, because its inventory rides the same slow public state — the leak moves one derivative earlier (Experiment I);

2. stacking such agents to the fixed point of mutual anticipation is algebraically one stronger such agent, and re-leaks harder while the model-space forecastable flow contracts to 0.237 (Experiment II);

3. a price-setter who absorbs the flow's impact exactly — the rational-expectations limit for this channel, verified to the identity — can close the leak only by deleting the de-leveraging dynamics that generate the tails, the leverage effect, and the crash asymmetry, scoring 1/10 (Experiment III).

Sign-efficiency and wildness are not separate dials: in a market moved by state-riding flows they are jointly produced, and no bolt-on layer of rationality — trading, iterated, or price-setting — buys the five missing events (E3, E4, E7, E8, E9). What the minimal market lacks is not a

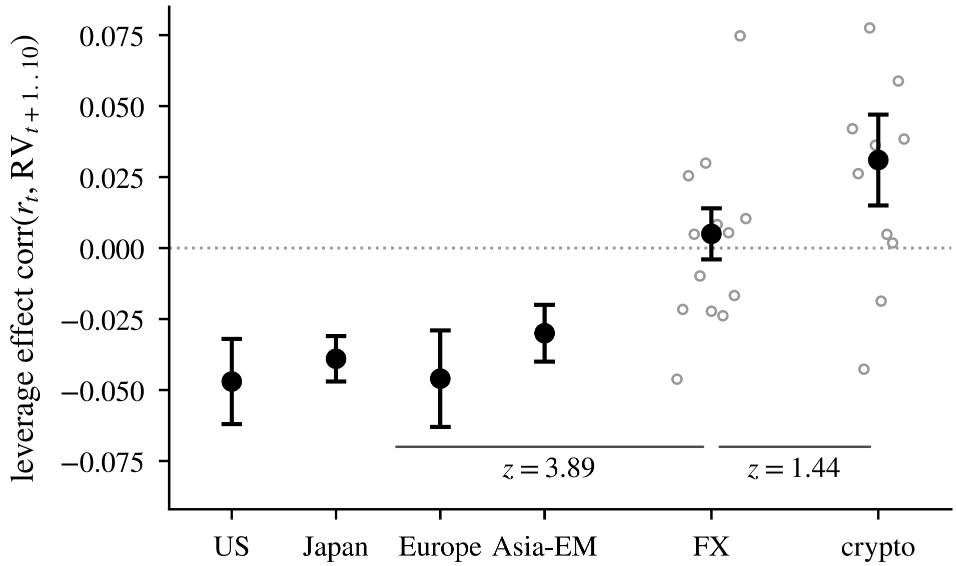

Figure 5. The leverage effect $\mathrm{corr}(r_t, \mathrm{RV}_{t+1..10})$ at three microstructure vertices. Filled points: cohort estimates ± 1 s.d. Open circles: the 13 individual FX crosses and 10 individual coins. Equities (four universes: US, Japan, Europe, Asia-EM) sit at -0.030 to -0.047; FX at $+0.005 \pm 0.009$ (statistically zero, $z = 0.56$); crypto at $+0.031 \pm 0.016$. The equity–FX separation is resolved ($z = 3.89$); the FX–crypto separation is not ($z = 1.44$).

smarter agent but a different kind of memory: the long-memory clock and the coupling arrow live upstream of the flows, in whatever generates them.

6.3 What is explicitly not claimed

The claim is scoped, and the scoping is part of the result. Explicitly *not* claimed: anything about real markets' generating process; anything beyond this model class (no learning agents, no heterogeneous information, no strategic equilibrium or rational-expectations solution concept); optimality of the agent designs — each is gate-verified to *function*, not to be optimal.

"Structural" here means: invariant over the searched class under the pre-registered budgets, with each searched member licensed by a mechanism gate proving it expresses its hypothesis. The claim would be falsified by an agent class outside the three — learning agents that condition on more than the public volatility state, or information-heterogeneous agents whose inventories do not ride a common public state — breaking the ceiling under the same battery and the same protocol. That is an open door, and we name it as such rather than hedge the claim through it.

7 Empirical counterpart: the microstructure triangle

The model mechanism makes the wildness channel a property of the *flow structure*: the leverage effect in the minimal market exists because a one-directional, state-riding de-leveraging flow exists, and dies when that flow's impact is removed (Figure 3). A weak but testable empirical counterpart: across real asset classes, the leverage effect should follow the microstructure that does or does not host such flows.

	equity cohort	FX	crypto
leverage effect	-0.0405 ± 0.0079	$+0.005 \pm 0.009$	$+0.031 \pm 0.016$
z vs. zero	-5.1	0.56	1.97
z vs. FX	3.89	—	1.44
instruments positive	0 of 4 markets	7 of 13 pairs	8 of 10 coins

Table 6. The leverage triangle. Equity cohort: four universes (US, Japan, Europe, Asia-EM), 2010–2026. FX: 13 liquid crosses, 2010–2026. Crypto: 10 major coins, 2017–2026. Each z vs. zero is the cohort mean divided by its cross-sectional dispersion (the $\pm$ values in the first row). The ordering $-0.041 < +0.005 < +0.031$ is monotone in point estimates; the equity edge is statistically certified, the FX–crypto edge is inside one-sided noise.

Table 6 and Figure 5 give the measurement, using an estimator battery identical across the three vertices (realized variance from daily OHLC ranges via Garman–Klass (Garman and Klass, 1980)). The classical readings of the equity effect — firm capital structure (Black, 1976) and volatility feedback through the equity risk premium (Bekaert and Wu, 2000) — are equity-specific mechanisms, which is exactly what makes the cross-venue comparison informative: a currency is not a levered claim and carries no equity premium, so whatever asymmetry survives outside equities must come from the venue's flow structure. Equity markets — where financial leverage and institutional de-risking cascades live — show the textbook negative effect in all four universes (-0.030 to -0.047). FX — an institutional dealer market hosting neither de-risking cascades nor a dominant retail-momentum dynamic — is statistically zero ($+0.005$, z vs. zero $= 0.56$; 7 of 13 pairs positive, the symmetric scatter a true zero predicts). Crypto — no financial leverage, no institutional de-risking, a retail dynamic in which rallies rather than crashes spike volatility — sits at $+0.031$, with 8 of 10 coins individually positive and only BTC (-0.043) and ETH (-0.019) retaining the equity-style sign.

Honesty about what is certified: the equity edge is resolved (one-sided $z = 3.89$ against FX, $z = 4.06$ against crypto), and FX is decisively zero; but the FX–crypto edge is $z = 1.44$, inside noise, and crypto itself clears zero at only $z \approx 2$ — so "crypto genuinely positive above an FX zero" is an ordering of point estimates, not a certified separation. The residual per-pair scatter in FX lies on the flight-to-quality axis: the three most negative crosses are all yen pairs (AUDJPY -0.046, GBPJPY -0.024, EURJPY -0.022 — yen strength is risk-off, so "down $\to$ vol up" is flight-to-quality wearing leverage's clothes), and the most positive is USDMXN ($+0.075$, the same flow with the opposite quote orientation).

The reading is licensed by a sign-recovery gate: on synthetic worlds with a known leverage sign (equity-negative, inverted-positive, symmetric-zero), the estimator recovers each with wide separation and its spurious-leverage bound is $|0.04|$, so a zero is read as a zero and an inversion as an inversion.

The positioning is deliberate: this triangle is *consistent with* the thesis — the wildness channel follows the flow structure of the venue, being negative exactly where forced de-risking lives, zero in the institutional market without it, and (in point estimate) positive where an opposite retail dynamic dominates — but it is not proof of it. No claim is made that the minimal market's specific mechanism generates the real effect.

8 Related work

Cont (2001) fixed the stylized-facts target list this battery operationalizes; our contribution to that program is calibration at both ends (real data 10/10, GBM exactly 3/10) so that partial scores are interpretable. The agent-based literature descending from Lux and Marchesi (1999) and surveyed by LeBaron (2006) established that fundamentalist–chartist interaction buys fat tails and clustering; our ablation reproduces that result (and its fragility: chartists without market makers destroy efficiency, scoring 1/10) but differs in asking the complement — which facts a flow-based market *cannot* buy at any rationality depth, established by pre-registered refutation rather than by exhibition.

Two adjacent traditions ask closely related questions with different instruments. In the heterogeneous-beliefs models of Brock and Hommes (1998), the sophistication of agents' expectations is a bifurcation parameter of a deterministic pricing map: making agents more forward-looking changes the dynamics qualitatively rather than monotonically improving them. Our K-ladder is a stochastic, capital-bounded cousin of that exercise, and it finds the same non-monotonicity in a measured quantity — the sign leak grows as anticipation deepens. The minority-game literature (Challet and Zhang, 1997) and the intermittent agent-based markets of Giardina and Bouchaud (2003) derive stylized facts from the statistical mechanics of adaptive agents competing over a scarce resource, with realistic regimes appearing near phase boundaries; our battery differs in scoring a fixed list of calibrated events rather than exhibiting a regime, but the joint-production finding rhymes with that literature's lesson that efficiency and excess volatility are properties of one collective state, not separate features.

The paper's E9 event operationalizes the oldest target in the field. Samuelson (1965) proved that properly anticipated prices fluctuate randomly; Fama (1970) organized the empirical program around informational efficiency; and Grossman and Stiglitz (1980) showed that full informational efficiency is impossible when information is costly — an equilibrium amount of inefficiency must survive to pay the agents who remove it. The joint-production result is a mechanical relative of that impossibility with the cost relocated: in this model class the residual sign information is not a fee paid to arbitrageurs but a by-product of the same state-riding flows that generate the volatility structure, and the three experiments show it cannot be traded, iterated, or quoted away without taking the wildness with it.

The price-impact literature of Bouchaud and coauthors (Bouchaud et al., 2009, 2018) underlies the paper's central object: the idea that order flow is the proximate mover of prices and that liquidity provision digests predictable flow. Our quote-skewing maker is a minimal formalization of perfect digestion — and Experiment III's finding that exact absorption deletes the volatility dynamics along with the leak is, in that vocabulary, a statement that the market's wildness *is* its undigested flow. Filimonov and Sornette (2012) measured near-critical reflexivity in markets via the Hawkes branching ratio; the measurement program behind this paper found that result replicates raw (branching ratio 0.68 across 48 US assets) but collapses to 0.25 — a no-self-excitation null — once events are deformed by the volatility clock, so most apparent reflexivity at the daily horizon is volatility clustering; the same deformation-first discipline shapes the battery's E8 event. Rough volatility (Gatheral et al., 2018) motivates the battery's long-memory clock event E4: the measurement program replicated $H \approx 0.10$ on 16 years of SPY range-based volatility, and E4 is calibrated so that exponential-memory processes (GJR-GARCH) fail it — which is exactly the event no flow configuration and no rationality layer ever bought, consistent with rough volatility's message that the clock's memory

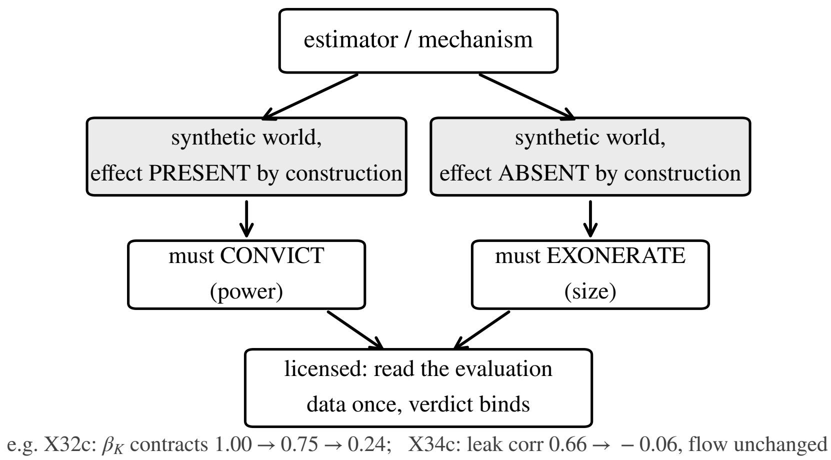

Figure 6. The gate pattern. Every estimator and every tested mechanism must, before any evaluation data is read, both detect a planted effect on a synthetic world where the effect exists (power) and stay silent on a world where it does not (size). The examples are this paper's: the fixed-point operator's contraction gate and the quote-skew maker's absorption gate.

is structural, not flow-generated.

9 Methods

9.1 Convict-and-exonerate gates

Every estimator in the program passes a gate before touching evaluation data: a test on synthetic data where the answer is known, proving both size (on a world where the effect is absent, the test does not fire) and power (on a world where the effect is present by construction, it does) (Figure 6). The battery's estimators were licensed this way before the first agent-based run, and — the step specific to this paper — so was every *tested mechanism*: the anticipator was proven to profit and damp flow impact on a toy world (Section 3), the fixed-point stack to contract forecastable flow (Section 4), the quote-skewing maker to absorb impact while leaving the flow untouched (Section 5). This is what makes the refutations evidential: when a licensed mechanism fails to move the battery, the failure is a property of the market, not of a broken implementation. At the time of writing the continuous-integration suite runs the numbered gates X1–X35 on every commit.

9.2 Pre-registration, budgets, and tie-breaks

"Pre-registered" here means: the design document — hypotheses, budgets, kill criteria — is committed to the repository before any evaluation run, and the version-control history is the timestamp; there is no external registry. Each experiment fixed, before any evaluation run: hypotheses (with one starred primary), the full parameter set, the evaluation protocol (8 seeds, IDs 100–107, $T = 6000$, majority vote per event, identical across all three experiments so ablations are paired seed-for-seed), a kill criterion, and a bounded tuning budget — one grid pass on disjoint tuning seeds (900–903), permitted only under a pre-declared trigger, with a deterministic pre-declared tie-break ("weakest intervention wins"). Amendments were permitted only before evaluation data was read, and each is recorded in the registration documents with its trigger. Two facts about this protocol did real work. First, the kill criteria fired three times and were honored three times. Second, the tie-break rule turned the tuning passes themselves into evidence: in all three experiments the battery's preferred setting was the weakest intervention available — the market scores best when the rational agent barely acts — which is the structural claim visible in the selection pattern before it is visible in the scores.

9.3 Reproducibility discipline

Three practices matter for the trustworthiness of small numeric differences. (1) *Byte-identity pins*: every new mechanism ships behind a flag, and a gate asserts that flag-off output is byte-identical to the pre-change simulator, so the calibration (SPY 10/10, GBM 3/10) and all previously published rows are protected across the entire arc; Experiment II's K=1 reproduces Experiment I's published configuration exactly, and Experiment III's λ_Q=0 reproduces both. (2) *Causality gates*: tampering with future inputs must leave the trade or quote prefix bit-identical (and the tail must differ, proving the mechanism actually reads the tape). (3) *Architecture-portable assertions*: an early version of the pins asserted absolute byte hashes of floating-point output, which held on one CPU architecture and failed on another by units in the last place; the discipline that survives is to assert in-process equivalence and architecture-stable invariants rather than absolute hashes.

Separately from this paper's claims: performance claims elsewhere in the same research program are governed by a trial ledger and reported as deflated Sharpe ratios (Bailey and López de Prado, 2014) beside a combinatorially cross-validated probability of backtest overfitting (Bailey et al., 2017); the program's standing rule is that its PBO of 0.45 is restated beside any Sharpe claim. This paper makes mechanism claims, not performance claims, but the same accounting culture — count the shots, publish the misses — produced it.

9.4 Code and data availability

Every number in this paper is regenerable from the public repository. The research JSONs consumed by the figures and tables — `decathlon.json`, `decathlon2.json`, `decathlon3.json`, `decathlon4.json`, `fx.json` and `crypto.json`, all under `research/` — are versioned in the repository and rebuilt by `run_research.py`; the battery code is identical for real data and simulated output; the gate suite runs in continuous integration on every commit, including the byte-identity pins that protect every published row in this paper; and all six figures are generated from the research JSONs by `docs/paper/make_figures.py` with no hand-entered values. The pre-registration documents, including the amendments and the closure clause quoted here, are versioned in `docs/design/`.

A The battery: exact criteria and calibration statistics

This appendix states each event's statistic and pass region exactly as implemented in the repository (`kronos/decathlon.py`, function `battery`; the thresholds were fixed in the DESIGN8 registration and its calibration-phase amendments). Notation: r_t are daily returns; RV_t is the 5-day rolling mean of r^2 (the close-only volatility clock), $\sigma_t = \sqrt{\mathrm{RV}_t}$, and $z_t = r_t/\sigma_t$ the

#	Statistic	Pass region
E1	$\mathrm{AC}_1(r)$	$[-0.15,\ 0.05]$
E2	kurtosis(r)	$[4.5,\ 40]$
E3	$\mathrm{AC}_1(\|r\|)$ and $\mathrm{mean}_{k=5..20}\,\mathrm{AC}_k(\|r\|)$	≥ 0.12 and ≥ 0.05
E4	$\mathrm{AC}_8(w)$ (weekly clock level)	≥ 0.12
E5	$\mathrm{mean}_{\tau=1..10}\,\mathrm{corr}(r_t, \mathrm{RV}_{t+\tau})$	≤ -0.03
E6	kurtosis(z)	≤ 5
E7	skew(u) (weekly clock AR(1) innovations)	≥ -0.35
E8	$\mathrm{EP}(r)$, $\mathrm{EP}(z)$ (entropy production, bits)	$\mathrm{EP}(r)$ significant and $\mathrm{EP}(z) \leq 0.75\,\mathrm{EP}(r)$
E9	direction bits $I(\text{features}; \operatorname{sign} r_{t+1})$	$\leq$ 95th percentile of 120 shuffles
E10	$\#\{r < -2.5s\} \,/\, \#\{r > +2.5s\}$, $s = \mathrm{sd}(r)$	≥ 1.25

Table 7. Exact pass criteria, as implemented. E7's floor is calibrated against the log-χ^2 noise skew of the weekly clock (GBM innovations skew ≈ -0.65; real SPY -0.22). E8's entropy production symbolizes the series into 3 bins, measures 3-gram irreversibility, and tests against 120 block-reversal surrogates (random block offsets, blocks reversed with probability 1/2). E9's features are the Cartesian product of three tape observables — sign(r_t), the sign of 21-day momentum, and the volatility tercile — scored by discrete plug-in mutual information with Miller–Madow bias correction against a 120-permutation null; the program's continuous information channel uses the KSG estimator (Kraskov et al., 2004), but E9's features are discrete by construction. The volatility tercile is cut on full-sample quantiles: E9 is a measurement of information content, computed identically on real and simulated series, not a trading rule.

deformed (one-clock) return. The weekly clock is built from *non-overlapping* 5-day means w_j of $\log r^2$ — raw weekly differences carry an MA(1) artifact on which even GBM's $|\Delta w|$ autocorrelates, so E4 uses the level and E7 uses AR(1) innovations $u_j = w_{j+1} - \bar{w}(1-\phi) - \phi w_j$ with ϕ the fitted AR(1) coefficient. $\mathrm{AC}_k(\cdot)$ is the lag-k autocorrelation. A configuration passes an event if a majority of its 8 evaluation seeds pass.

Two calibration facts anchor the thresholds. GBM's weekly-clock innovation skew (-0.651) is the log-χ^2 noise floor E7 is calibrated against; real volatility's up-jumps cancel most of it (SPY -0.221). And the E4 level-autocorrelation criterion is what a GJR-GARCH process fails (clock $\mathrm{AC}_8 \approx 0.06$ against the 0.12 bar): exponential memory decays too fast at eight weeks, which is why E4 is the battery's long-memory event rather than any Hurst point estimate (the close-only Hurst proxy is biased and is recorded as informational only).

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Statistic	Event	SPY (real)	GBM (G)	FCVM
$\mathrm{AC}_1(r)$	E1	−0.103	0.001	0.036
kurtosis(r)	E2	15.09	3.04	8.79
$\mathrm{AC}_1(\lvert r\rvert)$	E3	0.307	−0.001	0.377
mean $\mathrm{AC}_{5..20}(\lvert r\rvert)$	E3	0.218	0.002	−0.050
$\mathrm{AC}_8(w)$	E4	0.186	−0.018	−0.003
leverage	E5	−0.113	0.007	−0.125
kurtosis(z)	E6	2.33	2.13	2.20
skew(u)	E7	−0.221	−0.651	−0.408
EP(r) bits	E8	0.0211	0.0002	0.0929
EP(z) bits	E8	0.0141	0.0006	0.1392
direction bits	E9	0.0029	0.0001	0.0184
tail asymmetry	E10	1.56	0.91	39.96

Table 8. The full statistic vector at the battery's calibration ends (SPY 10/10, GBM 3/10) and at the flow-only ceiling (FCVM, 5/10); GBM and FCVM entries are medians over the 8 evaluation seeds. The table shows *how* each score decomposes: GBM passes only E1, E6, E9; FCVM matches SPY's magnitudes on E1/E2/E5/E10 but has no clock memory ($\mathrm{AC}_8(w) \approx 0$ against SPY's 0.186), fails E8 on the ratio criterion ($\mathrm{EP}(z) > 0.75\,\mathrm{EP}(r)$: deformation does not reduce its entropy production), and leaks 6× SPY's direction bits. Its E7 skew (−0.408) sits marginally below the −0.35 floor. SPY's EP(z) is significant but 33% below EP(r), meeting the ratio criterion.

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